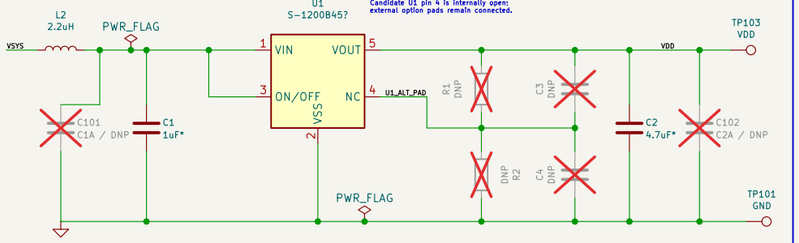
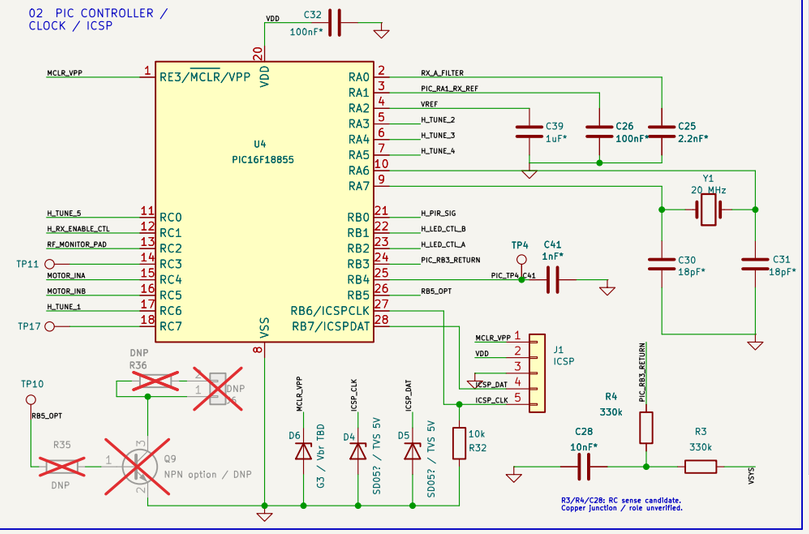
01 POWER / INPUT FILTER

U1: S-1200B45 candidate: VDD measured 4.5 V.



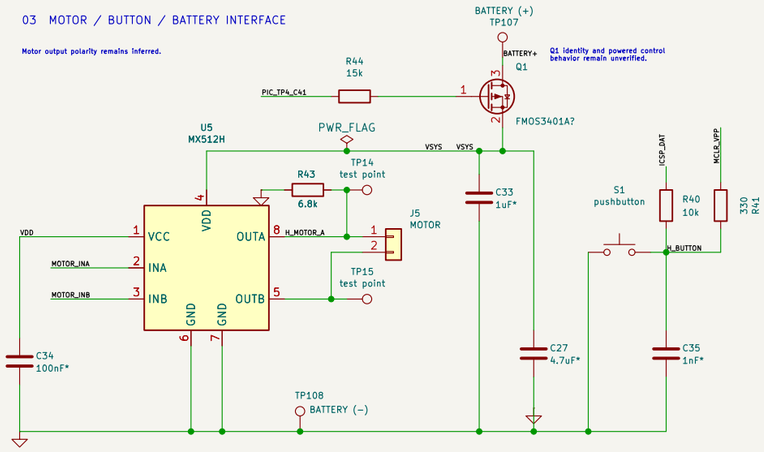
02 PIC CONTROLLER / CLOCK / ICSP



03 MOTOR / BUTTON / BATTERY INTERFACE

Motor output polarity remains inferred.

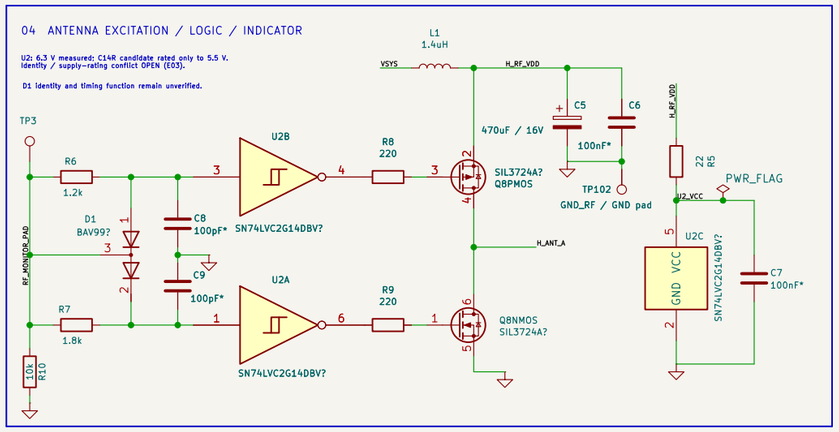
Q1 identity and powered control behavior remain unverified.



04 ANTENNA EXCITATION / LOGIC / INDICATOR

U2: 6.3 V measured; C14R candidate rated only to 5.5 V. Identity / supply-rating conflict OPEN (E03).

D1 identity and timing function remain unverified.

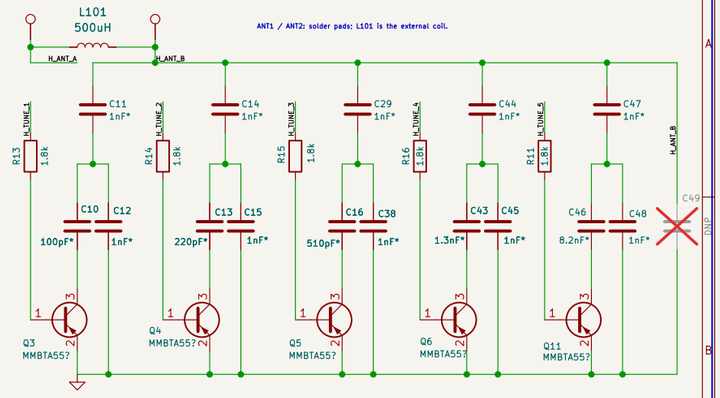


06 SWITCHED ANTENNA CAPACITANCE

Tuning assumes 500 uH and 134.2 kHz; switching weights/order remain unverified.

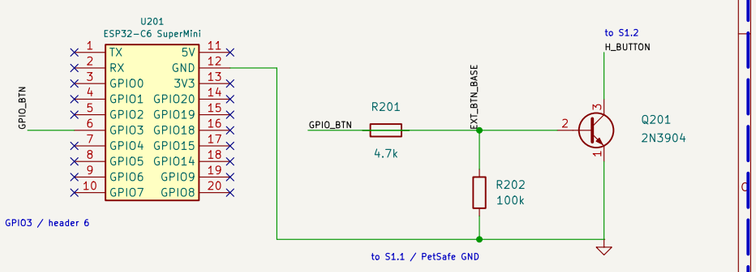
Q3 diode readings dispute the PNP pinout. Q3/Q4/Q5/Q6/Q11 identities remain unverified.

ANT1 / ANTENNA A ANT2 / ANTENNA B



EXTERNAL ESP32 BUTTON BRIDGE — PROPOSED MODIFICATION

USB power only; share GND, keep positive supplies separate.

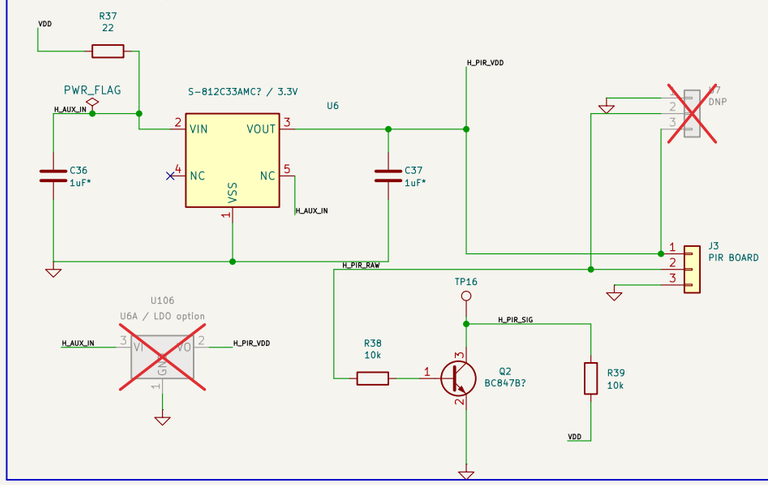


HIGH = brief press; LOW = release. Initial pulse ~200 ms (unverified timing).
Boot / reset must release. Never hold HIGH to keep unlocked.
Mode toggle only: no OPEN/CLOSE command or latch feedback.
Confirm owned TO-92 lead order and board startup before assembly.
USB powers U201 internally: 5V/3V3 header pins are deliberately unused.

07 PIR / AUXILIARY RAIL

U6: S-812C33 candidate; PIR supply measured 3.3 V.

U6A / U106: DNP regulator option. Symbol and ERC conflict retained intentionally.

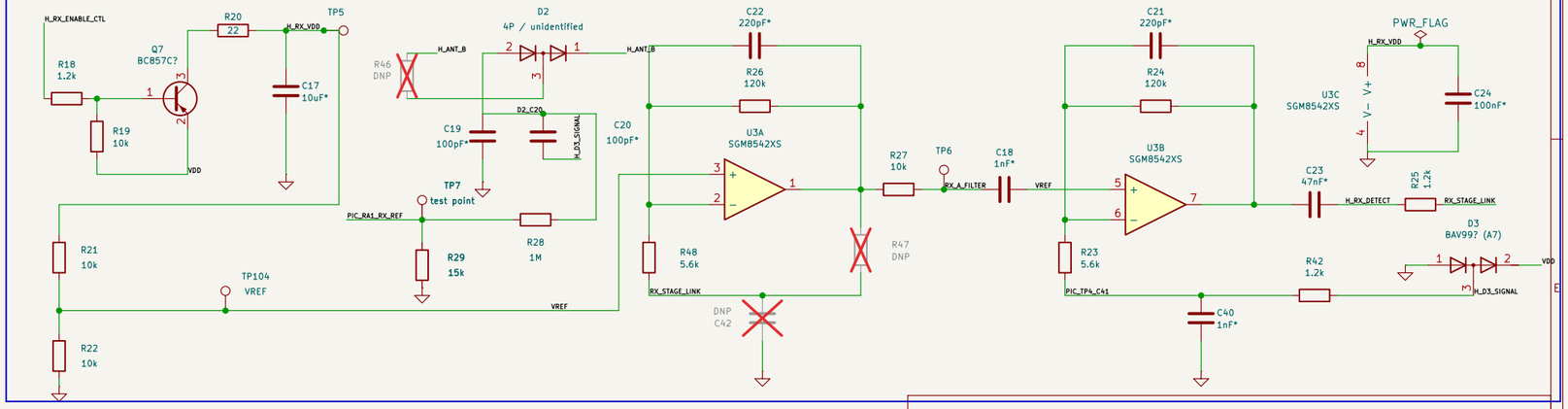


05 RF RECEIVE / GAIN / ENVELOPE

Receiver gain, bias behavior and Q7 switching remain unverified.

C42 / R47 / R46 are unpopulated options.

D2: measured series equivalent; identity unverified.



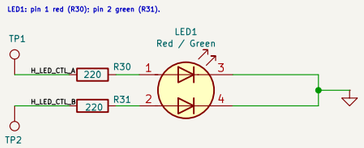
RECONSTRUCTION NOTES

Reviewed baseline; not fabrication-qualified.
* = capacitor estimate; ? = unverified identity / role.
Net names alone do not establish function.
DNP = not fitted; unknown routes are not NCs.

Pad mapping: evidence/pin_crosswalk.json
Open items: docs/OPEN_ITEMS.md
TP11 / TP17 are valid endpoints; possible hidden branches remain unproved.

Main board, external antenna and proposed bridge.
PIR internals, motor and firmware are outside scope.
Capacitor basis / ratings: docs/CAPACITOR_VALUE_REVIEW.html

LED1: pin 1 red (R30); pin 2 green (R31).



44 starred capacitor estimates; device functions remain qualified.

Owner layout | reviewed reconstruction baseline

Sheet: /
File: PetSafe_1001339.kicad_sch

Title: PetSafe 100-1339 R03 A | single-sheet reconstruction

Size: A3	Date: 2026-10-17	Rev: 0.9.52
KiCad E.D.A. 10.0.5		Id: 1/1